
GENERAL RELATIVITY

Tutorial problem set 10, 20.11.2020, with solutions

■ PROBLEM 27 Palatini formalism.

The Einstein equation can be derived by varying the action $S = S_{EH} + S_{matter}$ with respect to the metric, where the Einstein-Hilbert action is considered a functional of the metric only. That means we have expressed the connection in terms of the metric (Christoffel connection).

The Einstein equation can also be derived by treating the metric and connection as independent degrees of freedom, and varying the action with respect to them separately. This is known as the Palatini formalism. Here, the Einstein-Hilbert action is

$$S_{EH} = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} g^{\mu\nu} R_{\mu\nu}(\Gamma) , \quad (27.1)$$

and $R_{\mu\nu}$ depends on the (unspecified) connection only.

- (a) Show that the variation with respect to the metric leads to the Einstein equation

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu} . \quad (27.2)$$

Note that the Einstein tensor $G_{\mu\nu}$ is defined in terms of Ricci tensor, which in turn depends on the yet unspecified connection. In order for this equation to make sense we need to vary the action with respect to the connection.

In the remainder of this problem you will show that if we assume a torsion-less connection, $\Gamma^\alpha_{[\mu\nu]} = 0$, that the variational principle tells us it also has to be metric compatible, $\nabla_\alpha g^{\mu\nu} = 0$. This then leads to (27.2) being the standard Einstein equation for the metric.

- (b) What is the variation of the Ricci tensor with respect to the connection?
- (c) Utilizing (ordinary) partial integration show that the requirement that the Einstein-Hilbert action be stationary under variation with respect to the connection implies the following relation

$$0 = -\frac{1}{\sqrt{-g}} \partial_\alpha (\sqrt{-g} g^{\mu\nu}) + g^{\mu\nu} \Gamma^\sigma_{\sigma\alpha} - g^{\sigma\nu} \Gamma^\mu_{\sigma\alpha} - g^{\mu\sigma} \Gamma^\nu_{\alpha\sigma} + \delta^\nu_\alpha \left[\frac{1}{\sqrt{-g}} \partial_\sigma (\sqrt{-g} g^{\mu\sigma}) + \Gamma^\mu_{\sigma\rho} g^{\sigma\rho} \right] . \quad (27.3)$$

Note that we cannot use the covariant version of Stokes theorem, since it only works for Christoffel connection, and here the connection is general.

- (d) Show the following,

$$\frac{1}{\sqrt{-g}} \partial_\alpha \sqrt{-g} = -\frac{1}{2} g_{\rho\lambda} \partial_\alpha g^{\rho\lambda} . \quad (27.4)$$

- (e) Using (27.4) and the definition of the covariant derivative, rewrite all the partial derivatives in (27.3) in terms of covariant ones, and show that the condition (27.3) reduces to

$$0 = -\nabla_\alpha g^{\mu\nu} + \frac{1}{2}g^{\mu\nu}g_{\rho\sigma}\nabla_\alpha g^{\rho\sigma} + \delta_\alpha^\nu \left[\nabla_\lambda g^{\mu\lambda} - \frac{1}{2}g^{\mu\lambda}g_{\rho\sigma}\nabla_\lambda g^{\rho\sigma} \right]. \quad (27.5)$$

- (f) By appropriate contractions of (27.5) with the metric tensor show (27.5) implies

$$\nabla_\alpha g^{\mu\nu} = 0, \quad (27.6)$$

that is, the Palatini variational principle implies metric compatibility.

SOLUTION: (a) Firstly, we know that $T_{\mu\nu} = -(2/\sqrt{-g})\delta S_{matter}/\delta g^{\mu\nu}$ and that $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$. Next, upon observing that $R_{\mu\nu}(\Gamma)$ depends on the connection only yields to the following variation of the action (27.1)

$$\delta_g S_{EH}[g, \Gamma] = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} \delta g^{\mu\nu} \left[R_{\mu\nu}(\Gamma) - \frac{1}{2}g_{\mu\nu}g^{\alpha\beta}R_{\alpha\beta}(\Gamma) \right]. \quad (27.7)$$

Combining this with

$$\delta_g S_{matter} = \int d^4x \frac{\delta S_{matter}}{\delta g^{\mu\nu}(x)} \delta g^{\mu\nu}(x) = \int d^4x \sqrt{-g} \left[-\frac{1}{2}T_{\mu\nu} \right] \delta g^{\mu\nu}. \quad (27.8)$$

gives (27.2)., where $G_{\mu\nu}(g, \Gamma) = R_{\mu\nu}(\Gamma) - \frac{1}{2}g_{\mu\nu}g^{\alpha\beta}R_{\alpha\beta}(\Gamma)$.

- (b) From the definition,

$$R^\alpha_{\mu\beta\nu}(\Gamma) = \partial_\beta \Gamma^\alpha_{\nu\mu} + \Gamma^\alpha_{\beta\lambda} \Gamma^\lambda_{\nu\mu} - \partial_\nu \Gamma^\alpha_{\beta\mu} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\beta\mu}. \quad (27.9)$$

Ricci tensor is then simply, $R_{\mu\nu}(\Gamma) = \delta^\beta_\alpha R^\alpha_{\mu\beta\nu}(\Gamma)$, and its variation is,

$$\delta R_{\mu\nu}(\Gamma) = \partial_\alpha \delta \Gamma^\alpha_{\nu\mu} + \delta \Gamma^\alpha_{\alpha\lambda} \Gamma^\lambda_{\nu\mu} + \Gamma^\alpha_{\alpha\lambda} \delta \Gamma^\lambda_{\nu\mu} - \partial_\nu \delta \Gamma^\alpha_{\alpha\mu} - \delta \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\alpha\mu} - \Gamma^\alpha_{\nu\lambda} \delta \Gamma^\lambda_{\alpha\mu}. \quad (27.10)$$

- (c) Varying the Hilbert-Einstein action in Palatini formalism (27.1) with respect to the connection gives,

$$\delta_\Gamma S_{EH} = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu}(\Gamma), \quad (27.11)$$

where $\delta R_{\mu\nu}(\Gamma)$ is given in (27.10). The partial derivatives that act on $\delta\Gamma$ can be integrated by parts to give,

$$\begin{aligned} \delta_\Gamma S_{EH} &= \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} \left[-\delta \Gamma^\alpha_{\nu\mu} \frac{1}{\sqrt{-g}} \partial_\alpha (\sqrt{-g} g^{\mu\nu}) + \delta \Gamma^\alpha_{\alpha\lambda} \Gamma^\lambda_{\beta\gamma} g^{\gamma\beta} + \delta \Gamma^\lambda_{\nu\mu} \Gamma^\alpha_{\alpha\lambda} g^{\mu\nu} \right. \\ &\quad \left. + \frac{1}{\sqrt{-g}} \delta \Gamma^\alpha_{\alpha\mu} \partial_\beta (\sqrt{-g} g^{\mu\beta}) - \delta \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\alpha\beta} g^{\beta\nu} - \delta \Gamma^\lambda_{\alpha\mu} \Gamma^\alpha_{\beta\lambda} g^{\mu\beta} \right] \\ &= \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} \delta \Gamma^\alpha_{\nu\mu} \left[-\frac{1}{\sqrt{-g}} \partial_\alpha (\sqrt{-g} g^{\mu\nu}) + \delta^\nu_\lambda \delta^\mu_\gamma \Gamma^\lambda_{\beta\gamma} g^{\gamma\beta} + \delta^\lambda_\alpha \Gamma^\beta_{\beta\lambda} g^{\mu\nu} \right. \\ &\quad \left. + \delta^\nu_\alpha \frac{1}{\sqrt{-g}} \partial_\beta (\sqrt{-g} g^{\mu\beta}) - \delta^\mu_\lambda \Gamma^\lambda_{\alpha\beta} g^{\beta\nu} - \delta^\lambda_\alpha \delta^\nu_\gamma \Gamma^\gamma_{\beta\lambda} g^{\mu\beta} \right], \quad (27.12) \end{aligned}$$

Assuming that the matter action does not depend on the connection (which is true for the scalar and vector field actions, but not for the fermionic action), setting the expression in the square brackets (27.12) to zero gives (27.3).

(d) Eq. (27.4) follows from the observation,

$$\frac{1}{\sqrt{-g}}\partial_\alpha\sqrt{-g} = \left[\Gamma^\beta_{\alpha\beta}\right]_{\text{Christoffel}} = \frac{1}{2}g^{\mu\nu}\partial_\alpha g_{\mu\nu} = -\frac{1}{2}g_{\mu\nu}\partial_\alpha g^{\mu\nu}. \quad (27.13)$$

(e) Upon rewriting

$$\begin{aligned} -\frac{1}{\sqrt{-g}}\partial_\alpha(\sqrt{-g}g^{\mu\nu}) &= \frac{1}{2}g^{\mu\nu}g_{\rho\lambda}\partial_\alpha g^{\rho\lambda} - \partial_\alpha g^{\mu\nu}, \\ \delta^\nu_\alpha \frac{1}{\sqrt{-g}}\partial_\sigma(\sqrt{-g}g^{\mu\sigma}) &= -\delta^\nu_\alpha \frac{1}{2}g^{\mu\sigma}g_{\rho\lambda}\partial_\sigma g^{\rho\lambda} + \delta^\nu_\alpha \partial_\sigma g^{\mu\sigma}. \end{aligned} \quad (27.14)$$

and converting partial derivatives into covariant derivatives, Eq. (27.3) can be rewritten as,

$$0 = -\nabla_\alpha g^{\mu\nu} + \frac{1}{2}g^{\mu\nu}g_{\rho\sigma}\nabla_\alpha g^{\rho\sigma} + \delta^\nu_\alpha \left[\nabla_\lambda g^{\mu\lambda} - \frac{1}{2}g^{\mu\lambda}g_{\rho\sigma}\nabla_\lambda g^{\rho\sigma} \right] + 2\Gamma^\rho_{[\rho\alpha]}g^{\mu\nu} + 2\Gamma^\mu_{[\alpha\beta]}g^{\beta\nu} + \delta^\nu_\alpha \Gamma^\rho_{[\beta\rho]}g^{\beta\mu}, \quad (27.15)$$

such that when space-time torsion vanishes, $T^\mu_{\alpha\beta} = 2\Gamma^\mu_{[\alpha\beta]} = 0$, one obtains (27.5).

(f) Firstly, multiplying (27.5) by $g_{\mu\nu}$ gives,

$$g_{\rho\sigma}\nabla_\alpha g^{\rho\sigma} = -2g_{\rho\alpha}\nabla_\sigma g^{\sigma\rho}. \quad (27.16)$$

Inserting this into (27.5) yields,

$$0 = -\nabla_\alpha g^{\mu\nu} + g^{\mu\nu}g_{\rho\alpha}\nabla_\sigma g^{\rho\sigma} + 2\delta^\nu_\alpha \nabla_\lambda g^{\mu\lambda}. \quad (27.17)$$

This can be contracted with δ^α_μ to give,

$$\nabla_\rho g^{\rho\nu} = 0, \quad (27.18)$$

which – when inserted into (27.17) – results in,

$$\nabla_\alpha g^{\mu\nu} = 0, \quad (27.19)$$

thus completing the proof. What we have learned in this exercise is that, if matter does not couple to the connection (*i.e.* when space-time torsion is not sourced by matter), then the Hilbert-Einstein and Palatini formulation of gravity are equivalent. However, fermions *do* source space-time torsion, implying that in general the Palatini and Hilbert-Einstein formulation of gravity are inequivalent.

■ PROBLEM 28 Energy conditions.

(a) The action for the scalar field is given by

$$S_\phi = \int d^4x \sqrt{-g} \left(-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi) \right), \quad (28.1)$$

where $V(\phi)$ is the potential. Under the assumption that $V(\phi) > 0$ and $g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi < 0$ show that the scalar field energy-momentum tensor satisfies the *dominant energy* condition.

(b) The action for the electromagnetic field is given by

$$S_{EM} = \int d^4x \sqrt{-g} \left(-\frac{1}{4} g^{\mu\alpha} g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} \right). \quad (28.2)$$

Show that the energy momentum tensor of the electromagnetic field satisfies the dominant energy condition.

SOLUTION: (a) The dominant energy condition (DEC) for a perfect fluid ($c = 1$),

$$T_{\mu\nu} = (\mathcal{P} + \rho) u_\mu u_\nu + g_{\mu\nu} \mathcal{P}, \quad (28.3)$$

states that $\rho \geq |\mathcal{P}|$ (DEC).

A scalar field energy momentum tensor,

$$T_{\mu\nu}^\phi = (\partial_\mu \phi)(\partial_\nu \phi) + g_{\mu\nu} \mathcal{L}_\phi, \quad \mathcal{L}_\phi = -\frac{1}{2}(\partial\phi)^2 - V(\phi), \quad (\partial\phi)^2 \equiv g^{\alpha\beta}(\partial_\alpha \phi)(\partial_\beta \phi), \quad (28.4)$$

can be brought into the perfect fluid form (28.3), with the identification,

$$u_\mu^\phi = \frac{\partial_\mu \phi}{\sqrt{-(\partial\phi)^2}}, \quad \rho_\phi = -\frac{1}{2}(\partial\phi)^2 + V(\phi), \quad \mathcal{P}_\phi = \mathcal{L}_\phi = -\frac{1}{2}(\partial\phi)^2 - V(\phi) \quad (28.5)$$

Now, by the conditions of the problem, $V > 0$ and $(\partial\phi)^2 < 0$, ρ_ϕ is a sum of two positive numbers, while the pressure is their difference, therefore it must be that, $\rho_\phi \geq |\mathcal{P}_\phi|$, the equality being satisfied only when $V(\phi) = 0$.

(b) Since the electromagnetic energy-momentum tensor cannot be converted into a perfect fluid form, we need the DEC conditions in their original form, namely DEC implies that WEC must be satisfied ($T_{\mu\nu} t^\mu t^\nu \geq 0$ for any timelike vectors) and in addition $T_{\mu\nu} t^\nu$ must be nonspacelike, *i.e.* $t^\mu T_{\mu\nu} g^{\nu\rho} T_{\rho\sigma} t^\sigma \leq 0$.

Let us start by writing the energy momentum tensor

$$T_{\mu\nu}^{\text{EM}} = F_{\mu\alpha} F_{\nu\beta} g^{\alpha\beta} + g_{\mu\nu} \mathcal{L}_{\text{EM}}, \quad \mathcal{L}_{\text{EM}} = -\frac{1}{4} g^{\rho\alpha} g^{\sigma\beta} F_{\rho\sigma} F_{\alpha\beta}. \quad (28.6)$$

Since energy conditions are scalar quantities, one can evaluate them in a convenient frame. Let us hence work in a Local Inertial Frame (LIF) in which $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$. To get to that frame, we have used 10 out of 16 degrees of freedom in coordinate transformations, so we still can use Lorentz transformations within LIF, and we can use them to bring the vector t^μ into the rest frame, *i.e.* $t^\mu = (1, 0, 0, 0)$. Working in this frame we have, $F_{0i} = E_i$, $F^{0i} = -E^i = -E_i$, $F_{ij} = \epsilon^{ijk} B_k$, $F^{ij} = \epsilon^{ijk} B_k$, such that

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} (F_{0i} F^{0i} + F_{i0} F^{i0} + F_{ij} F^{ij}) = \frac{1}{2} (\vec{E}^2 - \vec{B}^2), \quad (28.7)$$

where $\vec{E}^2 = \delta_{ij} E_i E_j$, $\vec{B}^2 = \delta_{ij} B_i B_j$. With this in mind, we can calculate the energy momentum tensor,

$$T_{00} = F_{0i} F_{0i} - \mathcal{L}_{\text{EM}} = \frac{1}{2} (\vec{E}^2 + \vec{B}^2) \quad (28.8)$$

$$T_{0i} = T_{i0} = F_{0j} F_{ij} = \epsilon^{ijk} E_j B_k = (\vec{E} \times \vec{B})_i \quad (28.9)$$

$$T_{ij} = -F_{i0} F_{j0} + F_{ik} F_{jk} + \delta_{ij} \mathcal{L}_{\text{EM}} = \delta_{ij} \frac{1}{2} (\vec{E}^2 + \vec{B}^2) - (E_i E_j + B_i B_j), \quad (28.10)$$

where for simplicity we dropped the superscript EM.

Now the first condition can be written as,

$$T_{\mu\nu}t^\mu t^\nu = T_{00} = \frac{1}{2} \left(\vec{E}^2 + \vec{B}^2 \right) \geq 0, \quad (28.11)$$

and it is thus satisfied. The second condition requires a contraction of, $T_{\mu\nu}t^\nu = T_{\mu 0}$ with T_0^μ , *i.e.*

$$\begin{aligned} T_{\mu\nu}t^\nu \eta^{\mu\rho} T_{\rho\sigma} t^\sigma &= -T_{00}^2 + T_{0i}T_{0i} \\ &= -\frac{1}{4} \left(\vec{E}^2 + \vec{B}^2 \right)^2 + \left(\vec{E} \times \vec{B} \right)^2 = -\frac{1}{4} \left(\vec{E}^2 - \vec{B}^2 \right)^2 - \left(\vec{E} \cdot \vec{B} \right)^2 \leq 0, \end{aligned} \quad (28.12)$$

where in the last equality we used, $\left(\vec{E} \times \vec{B} \right)^2 = \vec{E}^2 \vec{B}^2 - \left(\vec{E} \cdot \vec{B} \right)^2$. Since the inequalities (28.11–28.12) must hold in a general coordinate system, we conclude that DEC is satisfied for the electromagnetic energy-momentum tensor (28.6). Note that both conditions (28.11–28.12) can be expressed in terms of Lorentz invariant scalars, $F_{\mu\nu}F^{\mu\nu}$ and $*F_{\mu\nu}F^{\mu\nu}$, where $*F_{\mu\nu}$ is the Hodge dual of $F_{\mu\nu}$. Show that!